

Backend Architecture & System Design

Project: Rentora (PHP + MySQL MVP)

1. Core Logic & User Accounts

For this MVP, a complex multi-vendor dashboard is not required initially. However, to keep data organized and prevent spam, we will implement a **Hybrid Account System**:

- **Renters (I Want to Rent):** No login required. They can submit requests as "Guests" using just their email and phone number.
- **Equipment Owners (I Want to List):** Basic account creation is required behind the scenes. When they list an item, their email acts as their unique identifier so they can later edit or remove their listings.

2. Data Flow & Integration (How it connects)

Step 1: Frontend (HTML/JS) → User fills the form and clicks submit. JavaScript validates the data and sends an AJAX POST request.

Step 2: Backend (PHP API) → A PHP script (e.g., `process_listing.php`) receives the data, sanitizes it (to prevent hacking), and inserts it into the MySQL database.

Step 3: Webhook (Google Sheets/Airtable) → Instantly, the PHP script uses cURL to send a copy of this data to a webhook URL (Google Apps Script or Zapier), updating the client's spreadsheet.

Step 4: Notifications → PHP sends an automated confirmation email to the user and an alert to the admin.

3. MySQL Database Schema

The database will be lightweight and relational.

Table: users

Column	Type	Description
id	INT (PK)	Unique User ID
name	VARCHAR(100)	Full Name
email	VARCHAR(100)	Email (Unique)
phone	VARCHAR(20)	Contact Number

Table: equipments

Column	Type	Description
id	INT (PK)	Unique Item ID
owner_id	INT (FK)	Links to users.id
title	VARCHAR(150)	E.g., "Power Drill"
daily_rate	DECIMAL	Price per day
status	ENUM	'active', 'rented', 'inactive'

Table: rental_requests

Column	Type	Description
id	INT (PK)	Unique Request ID
renter_name	VARCHAR(100)	Guest Name

equipment_id	INT (FK)	Links to equipments.id (Optional for MVP)
start_date	DATE	Rental start date

4. Hosting & Infrastructure

Since the platform uses a standard PHP/MySQL stack, it will run seamlessly on standard web hosting (cPanel/DirectAdmin). **No complex external cloud servers are required**, as the hosting provider's native environment is fully sufficient for this MVP, keeping maintenance simple and costs low.